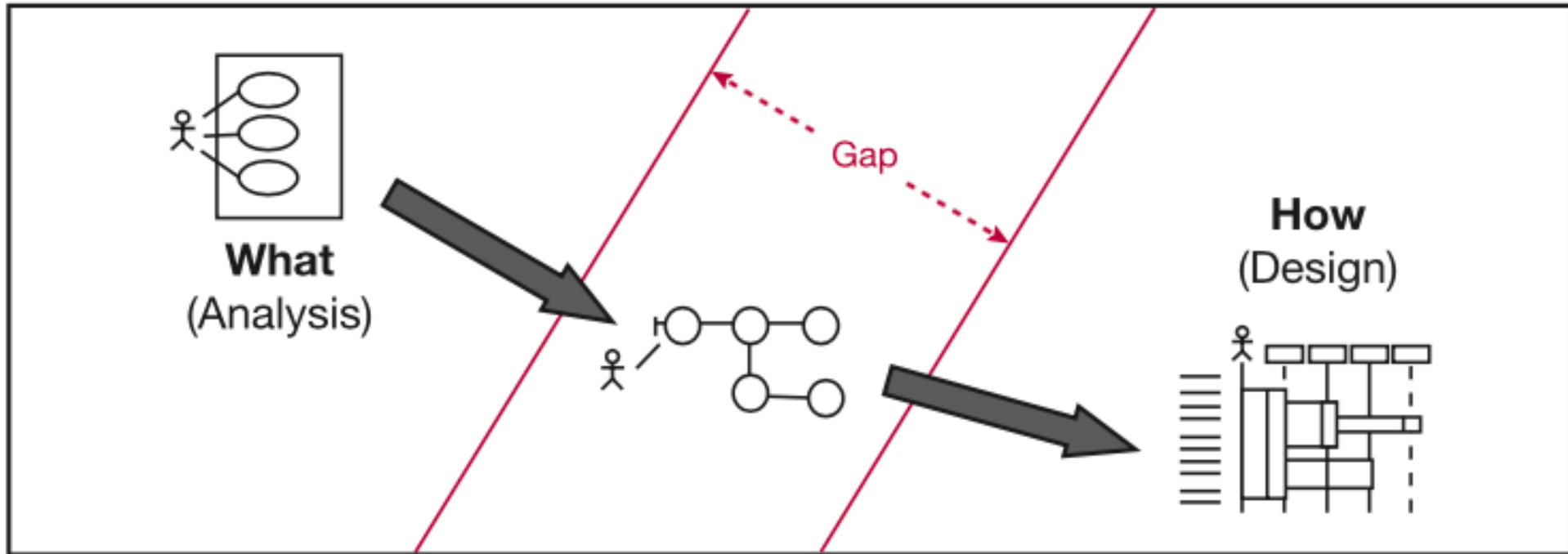


Robustness Analysis

Team Teaching ADBO 2025
Semester Genap 2024/2025
Prodi D4 Teknik Informatika
Jurusan Teknologi Informasi

What is Robustness Analysis? (1)

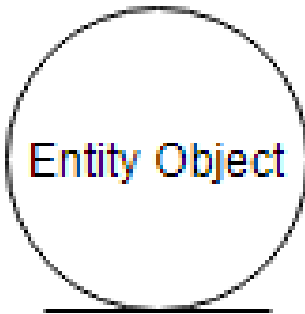


The tools to start making some **preliminary assumptions** about **design** and **technical architecture**.
It depicted with **Robustness Diagram**

Robustness Diagram

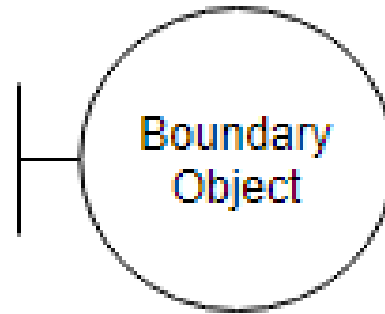
- Somewhat hybrid between class diagram and activity diagram
- Robustness diagram is an object picture of a use case
- Pictorial representation of the behavioral described by use case (but avoids showing which class is responsible to for which behavior)
- Read more like activity diagram

Robustness Diagram Symbol (Stereotypes)



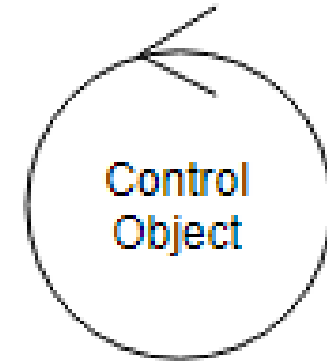
Entity Object

Object
representing
stored data



Interface / Boundary Object

Object at the
system
Interface

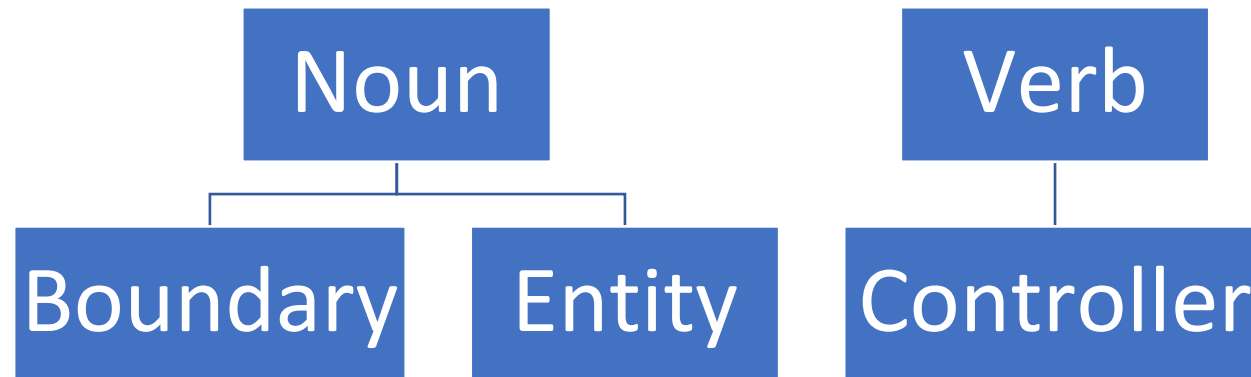


Control Object

Object representing
transfer of
information

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/robustness-analysis-tutorial/>

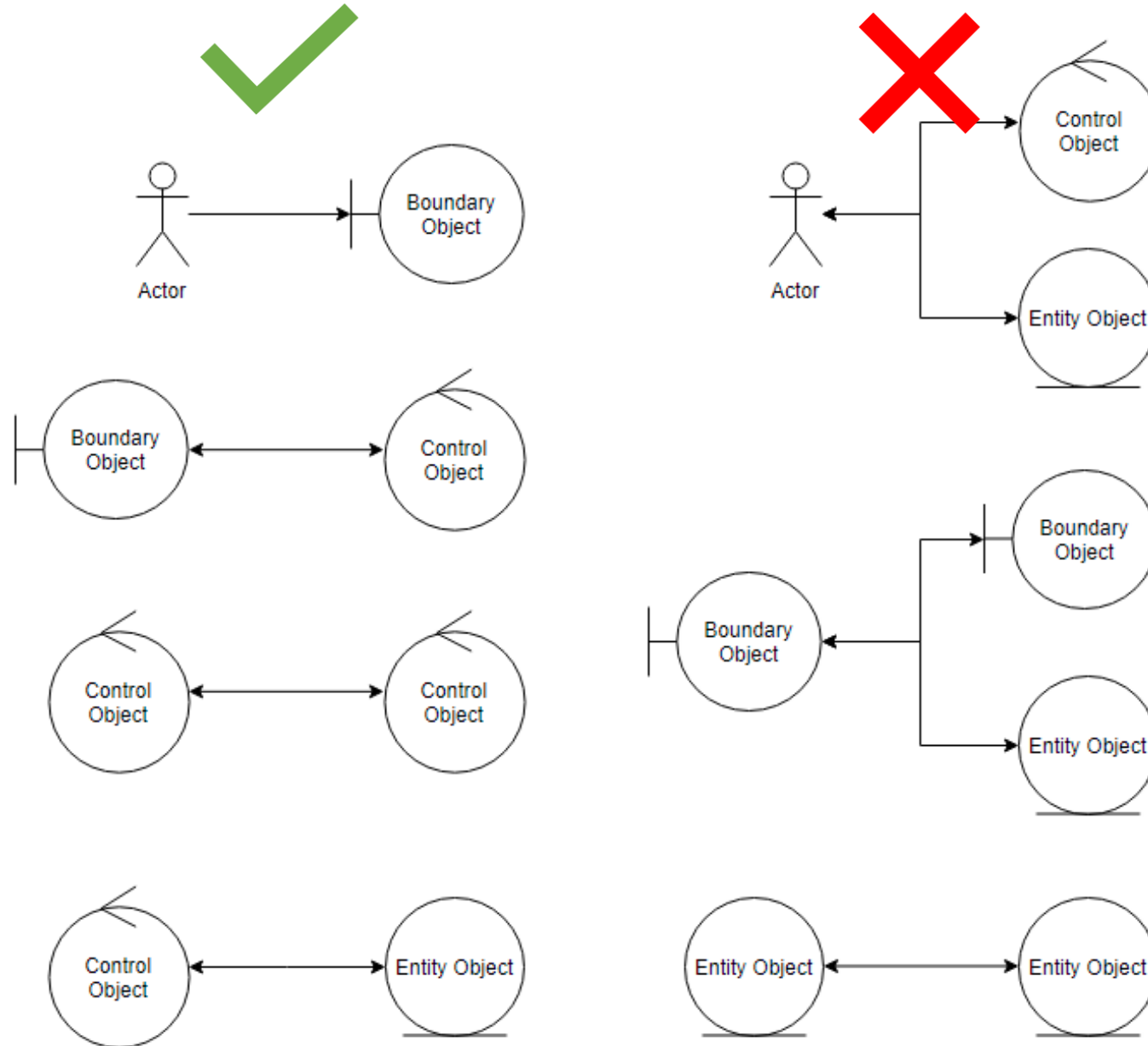
Robustness Diagram Communication Rules (1)



- **Nouns can talk to verbs (vice versa).** Therefore, boundary or entity can communicate with controller
- **Nouns cannot talk to other nouns.** Therefore, boundary cannot communicate to another boundary or entity (vice versa)
- **Verbs can talk to other verbs.** Therefore, controller able to communicate each other
- Additionally, **Actor** can **only be able** to communicate with **boundary object**

Robustness Diagram

Communication Rules (2)



<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/robustness-analysis-tutorial/>

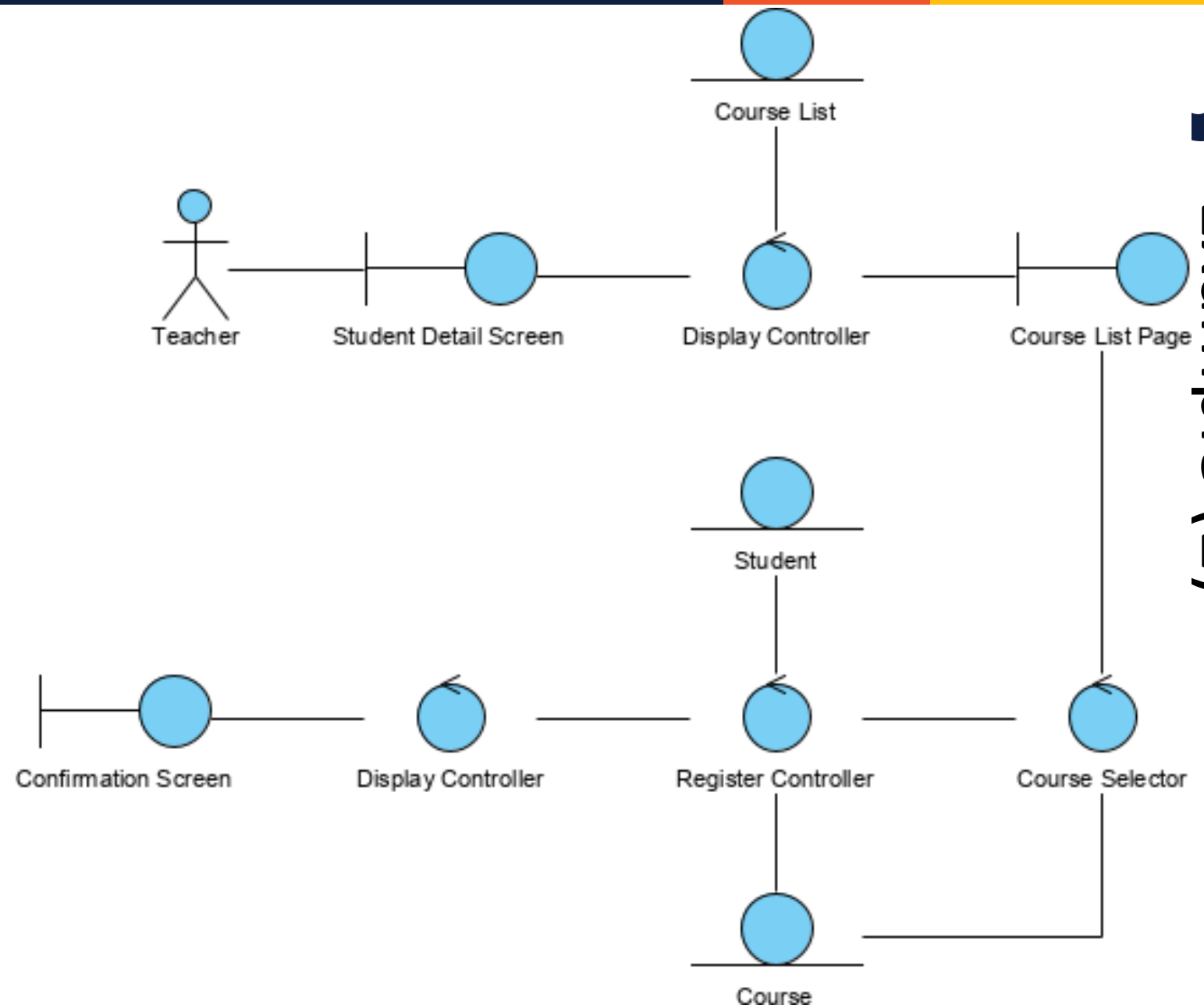
Example (1)



Use Case Description Basic Flow:

From the student detail page, the teacher clicks on the “Add courses” button and the system displays the list of courses. The teacher selects the name of a course and presses the “Register” button. The system registers the student for the course.

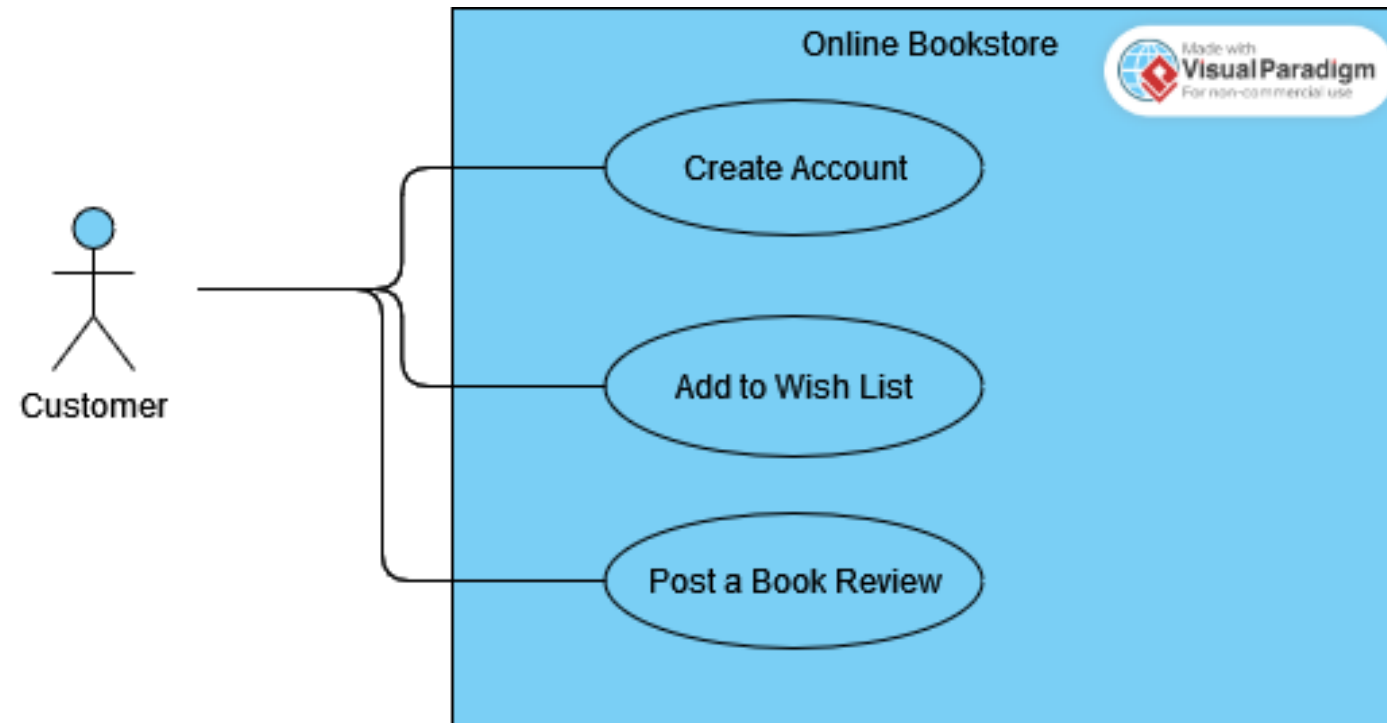
From the student detail page, the teacher clicks on the “Add courses” button and the system displays the list of courses. The teacher selects the name of a course and presses the “Register” button. The system registers the student for the course.



Back to our online bookstore

Refine, refine, and refine

Online Bookstore: The Use Case Diagram

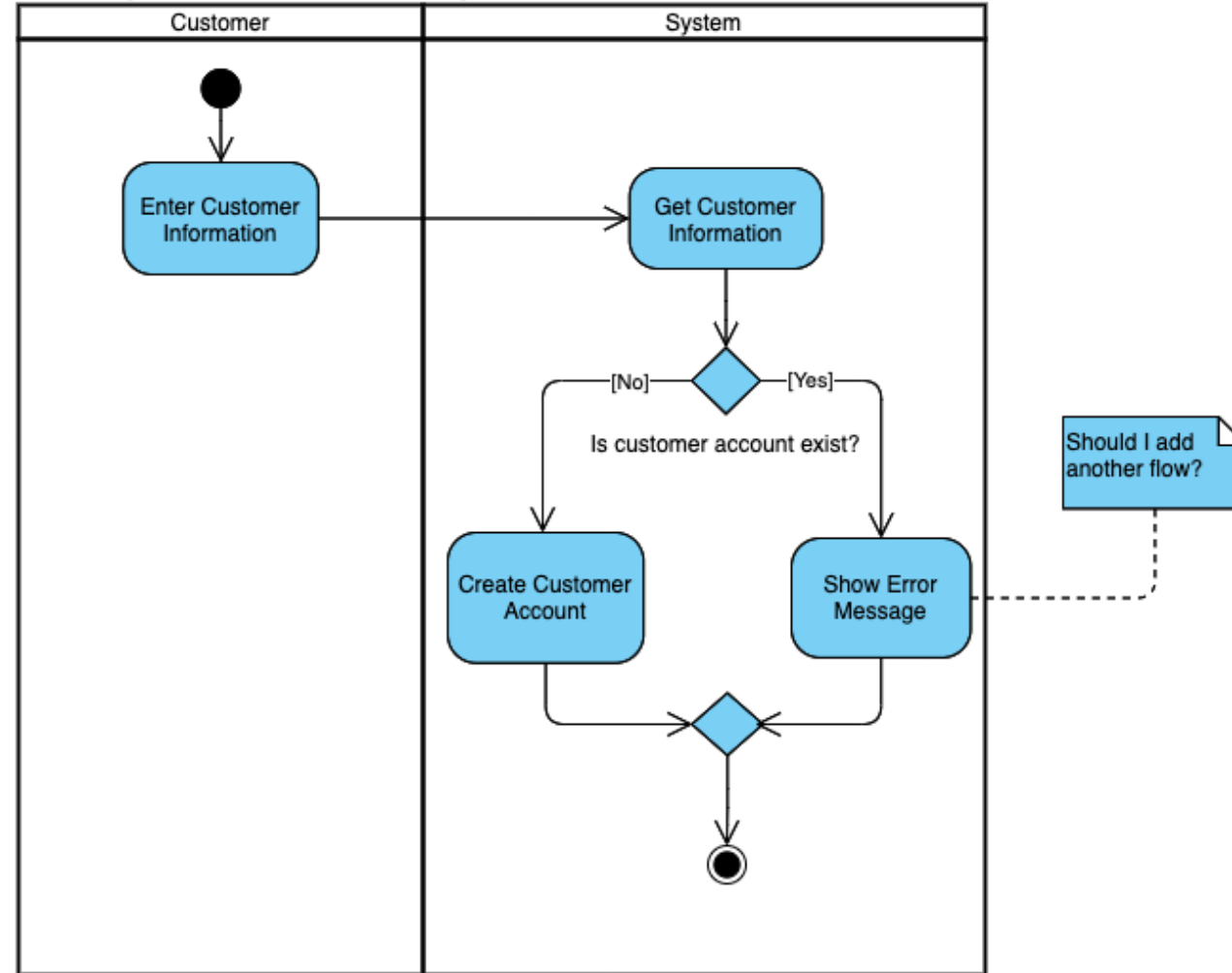


Online Bookstore: The Use Cases

- The **customer** enter id, name, address, email, and password, to **create** a **customer account** and then click create account button. The system looks up the **master account list**. If the customer doesn't exist, the system will add the **customer account** to **master account list**.
- The **customer** select a **book** and **adds** it to his **wish list**. The system display the updated wish list page and display the customer's **shopping chart**.
- The **customer** able to **post** a **book review**. The system will show the reviews on the book detail page.

The Activity Diagram of Create Account

Visual Paradigm Online Academic Partner Program



Visual Paradigm Online Academic Partner Program

The Use Case Description of Create Account (1)



Use Case Name: Create Account	ID: UC001	Importance Level: High
Primary Actor: Customer	Use Case Type: Detail, Essential	
Stakeholder and Interest: Customer – wants to create a new account in the online bookstore web application		
Brief Description: This use case describes how customer make an account		
Trigger: Customer click on register button at the online bookstore homepage		
Type: External		
Relationship:		

The Use Case Description of Create Account (2)



Normal Flow of Event:

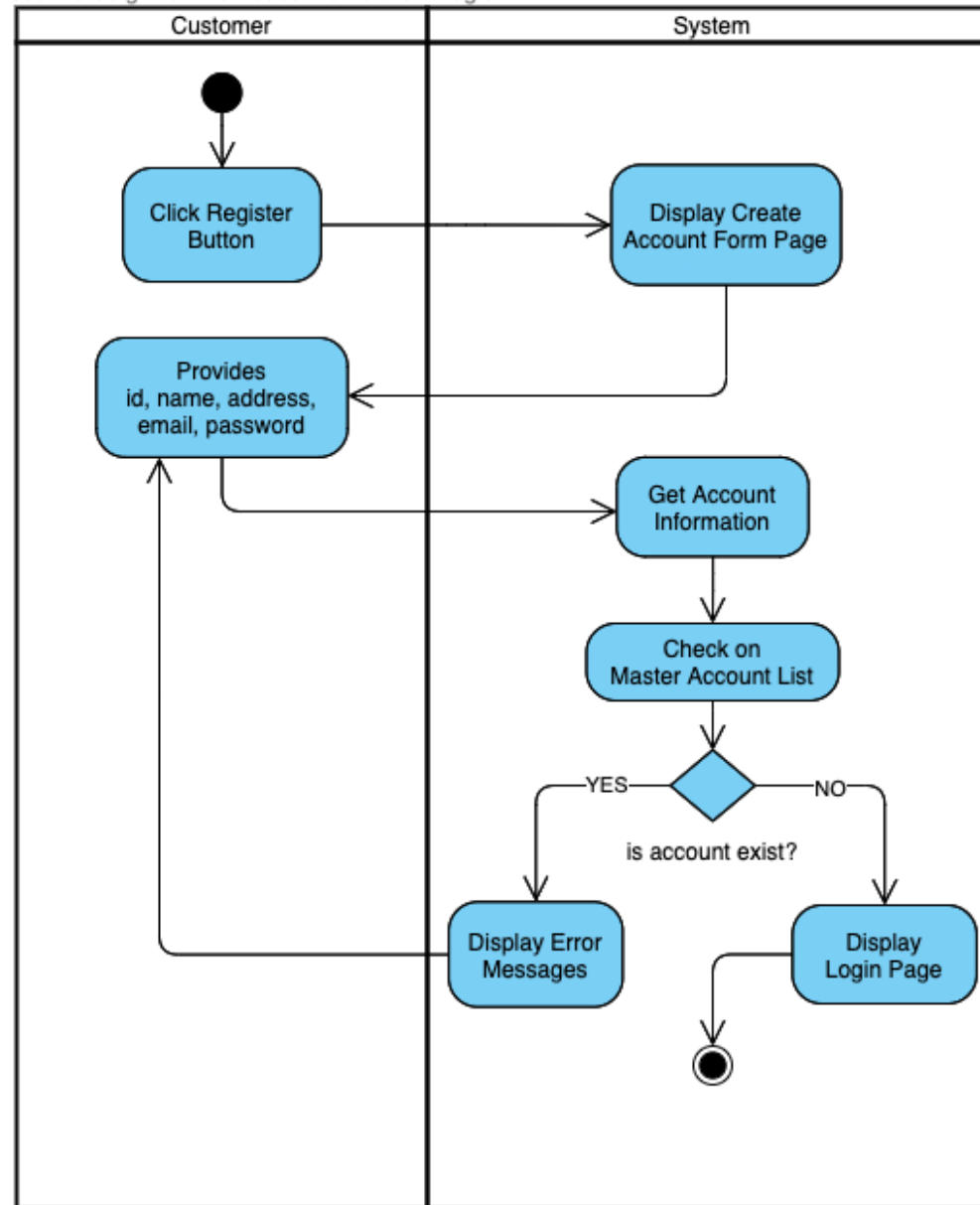
1. The customer opens the online bookstore
2. The customer clicks on register button
3. The customer provides id, name, address, email, and password
4. The customer clicks create account button
5. The system gets the customer information from point 3
6. The system checks master account list
7. If the account doesn't exist, the system creates a new account for the customer and display login page

Sub Flows:

Alternate/Exceptional Flow:

If account exist, the system displays an error message and redisplay the create account form page

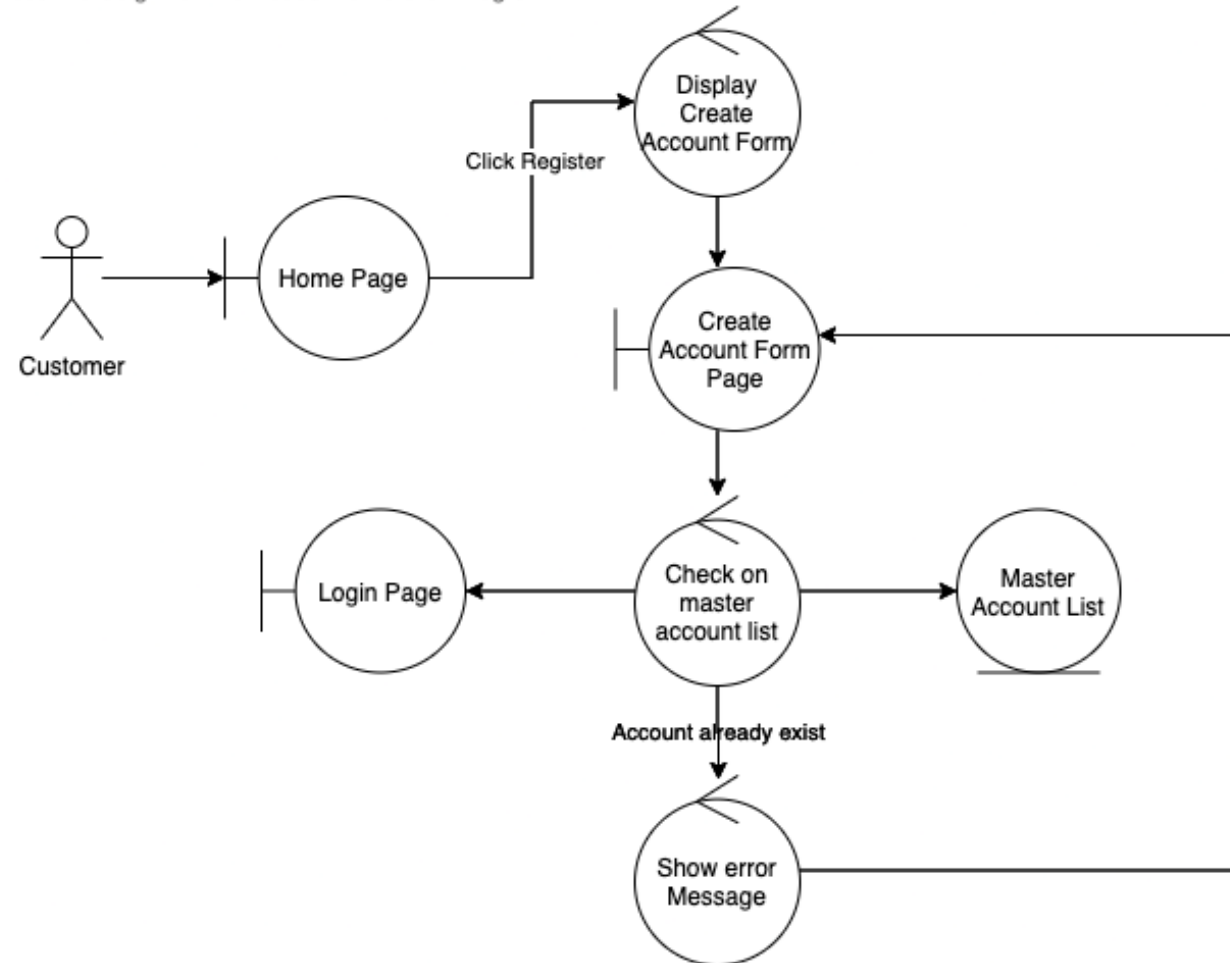
The Activity Diagram of Create Account [UPDATED]



Robustness Diagram of Create Account



Visual Paradigm Online Academic Partner Program



Visual Paradigm Online Academic Partner Program

